Demo Study D: Transformer Models for Satellite Link Quality Prediction

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SYNTHETIC DEMO DOCUMENT - fictional content for software testing. Not a real publication.

Abstract

This paper evaluates a Transformer model for predicting link quality (SNR) under rain. Results show that the Transformer does not improve over a simple LSTM in heavy rain conditions.

Introduction

Attention-based architectures have shown strong results in sequence modelling. We examine whether they offer benefits for satellite signal-to-noise ratio forecasting during heavy rain.

Data

The dataset is a public benchmark of SNR logs recorded at one Ku-band terminal over 8 months. The dataset is publicly available and the code is open source on GitHub.

Methodology

We train a Transformer encoder with 4 attention heads and compare it with an LSTM and a CNN. Metrics are RMSE, MAE, and R2. Feature importance was analysed with SHAP for interpretability.

Results

The Transformer obtains an RMSE of 1.12 dB, which is worse than the LSTM at 0.98 dB during heavy rain. The Transformer does not improve peak attenuation prediction and fails to generalize to a held-out month.

Limitations

Only one terminal was used, so site generalization was not tested. Inference cost of the Transformer is higher, but real-time performance was not evaluated.